

Cooperative Non-Orthogonal Multiple Access With Index Modulation for Air-Ground Multi-UAV Networks

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Abstract—Unmanned aerial vehicles (UAVs) serve as flexible aerial platforms, enriching air-ground communication networks in various ways. To support massive connectivity within limited time-frequency blocks, non-orthogonal multiple access (NOMA) is proposed to be integrated into UAV networks. However, a common issue associated with almost all NOMA schemes is the susceptibility to inter-user interference (IUI). Therefore, in this paper, we propose a multi-UAV cooperative system aided by NOMA with index modulation (IM), termed MCU-NOMA-IM, to improve the performance of air-ground networks by mitigating IUI and also avoiding the successive interference cancellation (SIC) decoding method that is prone to error floors. With MCU-NOMA-IM, the information bits pertaining to multiple UAVs are mapped into multiple dimensions, including the modulated symbols, subcarrier indices, and energy allocation patterns. To fully investigate the performance of MCU-NOMA-IM on air-ground networks, we consider scenarios in the presence of

three and four UAVs and derive upper-bounds for the bit error rates (BERs). In addition, we propose a multi-clustered-UAV cooperative system aided by NOMA with IM (MCCU-NOMA-IM), which groups closely located UAVs into several clusters to reduce the requirement for time resources. Simulation results demonstrate that both MCU-NOMA-IM and MCCU-NOMA-IM greatly outperform cooperative NOMA and non-cooperative NOMA-IM schemes, especially for distant UAVs when the signal-to-noise ratio is sufficiently high. Also, we show that the derived BER upper bounds are asymptotically tight.

Index Terms—Index modulation, non-orthogonal multiple access (NOMA), UAV-aided communications, air-ground networks, bit error rate (BER).

I. INTRODUCTION

AS a crucial component within integrated space-air-ground networks, unmanned aerial vehicles (UAVs) play a pivotal role in achieving extensive airspace coverage, with attributes encompassing long-range connectivity, high maneuverability, flexible deployment, and low latency. Over the last decade, UAVs have gathered substantial attention for their potential applications in various civil domains, including aerial inspection, precision agriculture, and traffic control [1], [2], [3], [4]. Among them, UAV-enabled communication is a particularly appealing paradigm, enriching the development of terrestrial communications by introducing aerial platforms [5], [6]. Currently, research on UAV-enabled communications typically follows two parallel streams: one focuses on quasi-static UAVs, and the other on mobile UAVs, depending on whether the UAV's high mobility needs to be exploited in specific application scenarios. In quasi-static scenarios, the UAV can remain at a fixed location for each communication period of interest. Conversely, mobile UAVs can fly closer to target points, with link directions evolving to improve the channel quality, and the Doppler effect caused by the mobility of the UAVs can be compensated via the schemes proposed in [7] and [8]. Moreover, owing to their versatility, UAVs can be employed to support wireless communications from several aspects: i) serving as an aerial base station (ABS); ii) serving as a relay node; and iii) serving as an aerial user. Some examples of UAV-enabled communications related to these applications are the following:

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- When considered as an ABS, one of the most attractive applications of UAVs is the ability to facilitate rapid communication recovery when terrestrial base stations (BSs) suffer severe damage after natural disasters such as earthquakes or typhoons. Also, UAVs used as ABSs can be utilized for enhancing network coverage and dynamically providing extra capacity on demand [6], [9], [10], [11].
- Using UAVs as smart relays is a major focus of 3GPP [12], [13], [14]. Due to their inherent mobility, UAVs are able to extend the communication range in several scenarios with limited infrastructure and fill coverage gaps [15]. Field tests in [16] have shown that using a UAV-based relay doubles the network throughput.
- As aerial users, UAVs can be utilized for data collection in various scenarios, including environmental monitoring. They can also be integrated into the existing cellular networks to facilitate air-ground communications [3]. The coexistence of ground and aerial users has been analyzed in the literature, even in the presence of severe air-ground interference [17], [18].

Like other communication systems, UAV-enabled communications are challenged by the availability of spectrum, as the allocated L-band and C-band are shared with other existing systems [19], [20]. Besides, the limited cruise time is a bottleneck for UAV applications, since the endurance duration of low-altitude deep-penetration UAVs is typically within one hour [21]. Therefore, how to support high-performance UAV-ground communications in a spectral and energy efficient manner is a key open problem [3], [17], [22]. Non-orthogonal multiple access (NOMA), a multiple access technology allowing multi-users to share the same time-frequency resource blocks, can boost the spectral efficiency (SE) and edge user throughput while reducing the transmission latency [23], [24]. As a highly efficient transmission technique, the integration of NOMA into UAV communication networks has the potential to provide benefits and solutions to design spectral and energy efficient UAV-aided communications. As for the UAV-ABS application, the authors of [25] considered a single UAV-ABS to communicate with two ground users utilizing NOMA, and they also studied the outage performance. A similar multiple-antenna-aided NOMA system was further introduced in [9]. Based on these works, the application of NOMA-aided UAV-ABS has been widely analyzed [10], [11]. Moreover, several research works evaluated the system design and optimization challenges associated with NOMA-enhanced UAV-aided relay networks. For example, a novel NOMA decode-and-forward UAV-based relay protocol has been developed to improve the data rate of cell-edge users in macrocell networks [26]. The authors of [27] considered the use of a NOMA-UAV as a relay to forward data from a remote BS to multiple ground users. As for cellular-enabled UAV communication networks, where the UAVs are considered to be aerial users, the NOMA technology was exploited in cooperative uplink transmissions from static UAVs to mitigate the air-ground interference [28]. Also, the authors of [29] studied an uplink NOMA system in the presence of a mobile UAV. In addition, the authors of [20] employed NOMA to tackle the spectrum scarcity problem in

full-duplex heterogeneous networks in the presence of multi-UAVs. These research works have confirmed that NOMA can assist UAV networks in simultaneously achieving better spectral and energy efficiency.

Notably, when adopting NOMA, cell-edge users typically suffer from poor communication performance due to long transmission distances and severe signal attenuation, which cannot be avoided by NOMA-UAV systems either. To overcome these limitations, the paradigm of cooperative NOMA (C-NOMA) was proposed in [30], wherein the cell-center user can serve as a relay to forward the data of cell-edge users. However, a common challenge that persists for almost all NOMA schemes is the inter-user interference (IUI) resulting from multiplexing the transmissions of many users in the power domain. Particularly, in practical scenarios where the channel cannot be perfectly estimated, the residual interference during the successive interference cancellation (SIC) process leads to a notable deterioration of the performance of NOMA systems. In [31], [32], [33], and [34], index modulation (IM) has been incorporated into C-NOMA systems, inspiring a series of IM-aided C-NOMA schemes. IM encodes additional information by utilizing, e.g., the antenna indices in space and subcarrier indices in frequency, resulting in a higher energy efficiency and lower implementation complexity [35], [36], [37]. Specifically, the authors of [31] and [33] proposed to encode the data of cell-edge users into antenna and subcarrier indices, introducing spatial modulation (SM) aided C-NOMA and orthogonal frequency division multiplexing with IM (OFDM-IM) aided C-NOMA, respectively. In these schemes, the data of multiple users can be independently carried by different information-bearing units, avoiding SIC and IUI. Besides, the performance of cell-edge users is further enhanced by employing IM, which reduces the error probability. Compared to SM-aided C-NOMA, OFDM-IM aided C-NOMA offers greater flexibility as it allows the frequency band to be divided into different sub-channels without the need for extra physical equipment. Additionally, OFDM-IM exhibits extensive scalability by introducing different modulation modes or extending the indexing to the energy domain. This has resulted in new modulation schemes, such as multiple-mode OFDM-IM (MM-OFDM-IM) and OFDM with index and composition modulation (OFDM-ICM) [38], [39], [40].

In this context, we introduce the concept of OFDM-IM aided C-NOMA into UAV networks for improving the spectral and energy efficiency of the transmission links between the UAVs and ground devices. Specifically, a cellular-connected UAV system is studied in this paper, where multiple UAVs communicate with one ground station (GS), and we analyze how NOMA and IM facilitate multi-access communications among the UAVs. In the considered system, a UAV has two roles: as an aerial user for data collection, and as a cooperative relay for forwarding information to distant users. Here, multiple UAVs must take turn between receiving and transmitting across separate time frames, which poses challenges for system synchronization. Therefore, we assume that the UAVs, e.g., balloon and rotary-wing UAVs, remain quasi-static for a given period of time to connect to the GS and support near UAVs from a certain altitude.

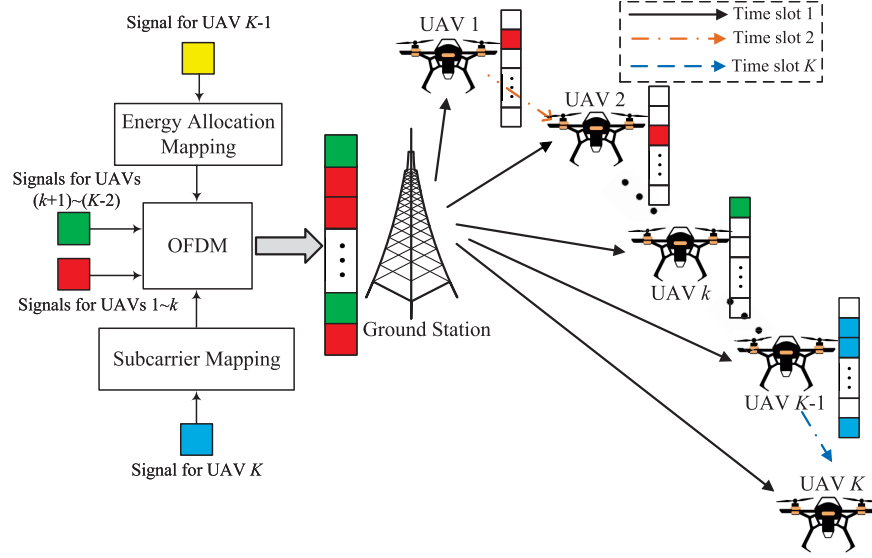


Fig. 1. System model of the considered MCU-NOMA-IM system with K UAVs.

The contributions of this paper can be summarized as follows:

- We propose a novel NOMA-IM scheme for enhancing multi-UAV-ground communications, which is termed multi-UAV cooperative system aided by NOMA with IM (MCU-NOMA-IM). Differently from existing works, this paper focuses on the error performance of NOMA-UAV networks. By encoding data onto multiple orthogonal dimensions, such as modulated symbols, subcarrier indices, and energy allocation patterns (EAPs), the signal associated with specific UAVs can be transmitted without inter-UAV interference. Moreover, with the collaborative efforts among UAVs, long-distance transmissions can be realized in an energy-efficient way.
- We consider the MCU-NOMA-IM scheme in the presence of three and four UAVs for illustration purposes, and investigate multi-UAV communication systems adopting NOMA-IM. The bit error rate (BER) of the proposed scheme is analyzed, and BER bounds are derived in closed-form expressions. Numerical simulations are utilized to validate the analysis, demonstrating that MCU-NOMA-IM has the potential to greatly outperform relevant benchmarks, such as multi-UAV-aided NOMA/C-NOMA, in terms of BER and SE.
- We further propose an enhanced scheme, which is termed multi-clustered-UAV cooperative system aided by NOMA-IM (MCCU-NOMA-IM), whereby near UAVs are grouped in a cluster to lower the requirements for time resources. The proposed MCCU-NOMA-IM scheme has the advantage of supporting more UAVs in a cluster, ensuring high-reliability, high-speed, and low-latency communications.

The remainder of this paper is organized as follows. Section II describes the system model of MCU-NOMA-IM. The error performance is analyzed in Section III. Then, the clustered scheme, i.e., MCCU-NOMA-IM, is proposed in Section IV, and simulation results are presented and discussed in Section V. Finally, Section VI concludes the paper.

Notations: Upper and lower case boldface letters denote matrices and column vectors, respectively. $(\cdot)^T$ and $(\cdot)^H$ represent the transpose and Hermitian transpose operations, respectively. $\mathbf{I}_M$ is the $M \times M$ identity matrix. $E\{\cdot\}$ denotes the expectation operation. $\|\cdot\|$ denotes the Frobenius norm. $C(\cdot, \cdot)$ returns the binomial coefficient. $\text{diag}\{\cdot\}$ denotes a diagonal matrix whose diagonal elements are given by the specified vector. $\det\{\cdot\}$ represents the determinant of a matrix. The probability of an event is denoted by $\Pr\{\cdot\}$.

II. SYSTEM MODEL

As illustrated in Fig. 1, we consider an MCU-NOMA-IM system with one GS and K UAVs, where all the devices are equipped with a single antenna. Due to the channel attenuation, each UAV is assumed to cooperate only with its nearest neighbor. We define the channel coefficients for the $\text{GS} \rightarrow \text{UAV } i$ link and the $\text{UAV } i \rightarrow \text{UAV } j$ link as $\mathbf{h}_{s_i}$ and $\mathbf{h}_{u_{ij}}$, respectively, where $i, j \in \{1, \dots, K\}$ and $i < j$. Besides, we assume a Nakagami- m fading channel for the cellular-connected UAV system, which allows us to control the severity of multi-path fading to emulate various scenarios, while retaining analytical tractability [41], [42]. Therefore, the entries in $\mathbf{h}_{s_i}$ and $\mathbf{h}_{u_{ij}}$ can be modeled as independent and identically distributed (i.i.d.) random variables, with their moduli following the Nakagami- m distribution with parameters $(m_\Delta, \Omega_\Delta)$, where m_Δ is an integer fading parameter and Ω_Δ is the local power for $\Delta \in \{s_i, u_{ij}\}$. Since the variance of the channel gain is inversely related to the communication distance, without loss of generality, we have $\Omega_{s_1} > \Omega_{s_2} > \dots > \Omega_{s_K}$ for the $\text{GS} \rightarrow \text{UAV } i$ link. In addition, we can observe from Fig. 1 that the entire transmission process for the considered system with K UAVs is completed in K time slots ($t_1 \sim t_K$), where t_1 denotes the broadcast phase and $t_2 \sim t_K$ represent the cooperative phase. To clearly illustrate the proposed scheme, we consider $K = 3$ and $K = 4$ in the following sections.

A. MCU-NOMA-IM With $K = 3$

As for the MCU-NOMA-IM system with $K = 3$, the whole communication protocol, consisting of the broadcast phase and the cooperative phase, is detailed next.

1) *Broadcast Phase:* During the broadcast phase, the GS transmits one IM vector with m information bits to all the UAVs via n subcarriers. These m bits can be decomposed into $m_1 \sim m_3$ bits, respectively, corresponding to UAV 1 ~ 3 ¹:

- $m_1 = k \log_2(M_1)$ bits are used to determine the symbol vector $\mathbf{s} = [s_1, \dots, s_v, \dots, s_k]$ to be transmitted on the selected k subcarriers, where s_v for $v \in 1, \dots, k$ is generated from the M_1 -ary phase-shift keying (PSK) or quadrature amplitude modulation (QAM) constellation set $\mathcal{S}_1$;
- $m_2 = (n - k) \log_2(M_2)$ bits are used to determine the symbol vector $\mathbf{a} = [a_1, \dots, a_l, \dots, a_{(n-k)}]$ to be transmitted on the $(n - k)$ unselected subcarriers, where a_l with $l \in \{1, \dots, (n - k)\}$ is generated from the M_2 -ary PSK or QAM constellation set $\mathcal{S}_2$;
- $m_3 = \lfloor \log_2 C(n, k) \rfloor$ bits are used to determine the subcarrier activation pattern (SAP) $\mathcal{I} = \{i_1, \dots, i_k\}$ with $\{i_\tau\}_{\tau=1}^k \in \{1, 2, \dots, n\}$.

Please note that to successfully split all modulated subcarriers into two parts and generate the corresponding index subsets, the constellation sets $\mathcal{S}_1$ and $\mathcal{S}_2$ need to be distinguishable. Based on the previous description, the transmitted signal vector in the frequency domain can be given by

$$\mathbf{x} = [s_1, s_2, a_1, \dots, s_k, \dots, a_{(n-k)}]^T. \quad (1)$$

The remaining processes for $\mathbf{x}$ at the transmitter are the same as those in classical OFDM, including the inverse fast Fourier transform (IFFT), cyclic prefix (CP) appending, and parallel-to-serial conversion. At the receiving side, after performing the fast Fourier transform (FFT) and CP removal, the received frequency-domain signal vectors at the three UAVs can be expressed as follows:

$$\mathbf{y}_{u_i}^{t_1} = \mathbf{H}_{s_i} \mathbf{x} + \mathbf{n}_{s_i}^{t_1}, \quad i \in \{1, \dots, K\}. \quad (2)$$

Since the channel matrix and the noise vector are frequently used in the following description, we define $\mathbf{H}_\Delta = \text{diag}\{\mathbf{h}_\Delta\}$, and $\mathbf{n}_\Delta^{t_i}$ as the additive Gaussian noise vectors with zero mean and variance N_0 in time slot t_i , where $\Delta \in \{s_i, u_{ij}\}$ corresponds to the communication link described previously for $i, j = \{1, \dots, K\}$ and $i < j$. Assuming perfect channel estimation at all the UAVs,² the transmitted signal for UAV 1 can be optimally detected using the maximum likelihood (ML) detection scheme, given by

$$[\hat{\mathcal{I}}_{u_1}^{t_1}, \hat{\mathbf{s}}_{u_1}^{t_1}, \hat{\mathbf{a}}_{u_1}^{t_1}] = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}} \|\mathbf{y}_{u_1}^{t_1} - \mathbf{H}_{s_1} \mathbf{x}\|^2, \quad (3)$$

¹In this paper, we assume that the importance of user information increases with distance, which is crucial in some scenarios, such as emergency communications. Therefore, in the proposed schemes, we send index or composition bits to the farthest UAV to ensure that the UAV with the most critical information operates optimally. Besides, we note that the proposed scheme can dynamically adjust the UAV performance by arranging different information-bearing units based on communication requirements.

²The assumption that the channel state information (CSI) can be perfectly estimated in air-ground networks is supported by the most recent channel estimation techniques [43], which have been widely adopted for UAV communication networks consisting of UAVs moving at low and intermediate speeds [44].

where $\hat{\mathcal{I}}_{u_1}^{t_1}$, $\hat{\mathbf{s}}_{u_1}^{t_1}$, and $\hat{\mathbf{a}}_{u_1}^{t_1}$ are the estimates of $\mathcal{I}$, $\mathbf{s}$, and $\mathbf{a}$ at UAV 1 in time slot t_1 , respectively. For cooperation purposes, UAVs 2 and 3 preserve the received signal vectors $\mathbf{y}_{u_2}^{t_1}$ and $\mathbf{y}_{u_3}^{t_1}$ for signal detection in the cooperation phase.

2) *Cooperation Phase:* In time slot t_2 , UAV 1 first reconstructs an IM signal vector using the estimated SAP $\hat{\mathcal{I}}_{u_1}^{t_1}$ and the estimated signal vector $\hat{\mathbf{a}}_{u_1}^{t_1}$, denoted as

$$\mathbf{f} = [0, \hat{a}_1, \dots, 0, \dots, \hat{a}_{(n-k)}]^T, \quad (4)$$

and then forwards $\mathbf{f}$ to UAV 2, yielding

$$\mathbf{y}_{u_2}^{t_2} = \mathbf{H}_{u_{12}} \mathbf{f} + \mathbf{n}_{u_{12}}^{t_2}. \quad (5)$$

By combining (2) and (5), the transmitted signal for UAV 2 can be estimated as follows:

$$[\hat{\mathcal{I}}_{u_2}^{t_2}, \hat{\mathbf{a}}_{u_2}^{t_2}] = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}} \left\{ \|\mathbf{y}_{u_2}^{t_1} - \mathbf{H}_{s_2} \mathbf{x}\|^2 + \|\mathbf{y}_{u_2}^{t_2} - \mathbf{H}_{u_{12}} \mathbf{f}\|^2 \right\}, \quad (6)$$

where $\hat{\mathbf{a}}_{u_2}^{t_2}$ is used to retrieve the information bits of UAV 2. During time slot t_3 , UAV 2 generates a space shift keying (SSK) signal using $\hat{\mathcal{I}}_{u_2}^{t_2}$, denoted as

$$\mathbf{q} = [0, 0, \dots, \underbrace{1, \dots, 1}_k, 0]^T, \quad (7)$$

and then transmits $\mathbf{q}$ to UAV 3. Therefore, the received signal at UAV 3 is given by

$$\mathbf{y}_{u_3}^{t_3} = \mathbf{H}_{u_{23}} \mathbf{q} + \mathbf{n}_{u_{23}}^{t_3}. \quad (8)$$

Similarly, the estimated SAP for UAV 3 can be obtained as

$$\hat{\mathcal{I}}_{u_3}^{t_3} = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}} \left\{ \|\mathbf{y}_{u_3}^{t_1} - \mathbf{H}_{s_3} \mathbf{x}\|^2 + \|\mathbf{y}_{u_3}^{t_3} - \mathbf{H}_{u_{23}} \mathbf{q}\|^2 \right\}. \quad (9)$$

The computational complexity of the optimal ML detector in UAV 1 ~ 3 in terms of complex multiplications is of order

$$\begin{aligned} \mathcal{C}_1 &= \left(n(M_1^k M_2^{n-k} 2^{\lfloor \log_2 C(n, k) \rfloor}) \right), \\ \mathcal{C}_2 &= \mathcal{C}_1 + (n - k)(M_2^{n-k} 2^{\lfloor \log_2 C(n, k) \rfloor}), \\ \mathcal{C}_3 &= \mathcal{C}_1 + k 2^{\lfloor \log_2 C(n, k) \rfloor}. \end{aligned}$$

Therefore, the computational complexity of ML detection for MCU-NOMA-IM with three UAVs is³ $\mathcal{C}_1 + \mathcal{C}_2 + \mathcal{C}_3$. Due to the similar calculation process, the analysis of the computational complexity of the ML detector is omitted in the following.

B. MCU-NOMA-IM With $K = 4$

In the previous subsection, we observe that three independent information-bearing units are assigned to the three UAVs: the modulation symbols transmitted on different subcarriers and the SAP. As a result, there is no interference between these UAVs. As the number of UAVs K increases, however, the design of the information mapping mechanism in the proposed MCU-NOMA-IM scheme needs to be generalized to avoid the IUI. To verify the feasibility of extending the proposed scheme to scenarios with multiple UAVs, this subsection discusses the communication process of the MCU-NOMA-IM system with four UAVs.

³The primary objective of this work is to maximize the performance gain of the cooperative NOMA-aided UAV framework. Therefore, despite its high complexity, the ML detector is employed to examine the proposed schemes. In practical applications, suboptimal detection methods, such as the log-likelihood ratio or compressive sensing methods, can be utilized to significantly reduce the complexity.

1) *Broadcast Phase*: We assume that m information bits are input into the MCU-NOMA-IM system, which can be decomposed into the following four parts:

- $m_1 + m_2 + m_4 = k \log_2(M_1) + (n - k) \log_2(M_2) + \lfloor \log_2 C(n, k) \rfloor$ represents the three components of the incoming bits, each of which is defined in Section II-A.1;
- $m_3 = \lfloor \log_2 C(I - 1, n - 1) \rfloor$ bits (also referred to composition bits) are used to determine the EAP $\zeta = \{l_1, \dots, l_n\}$ with $\{l_i\}_{i=1}^n \in \{1, 2, \dots, I - 1\}$, where I is the energy activation factor.

Here, m_i bits correspond to UAV i for $i = 1, \dots, K$. Accordingly, we find that a new information-bearing unit, the EAP, is introduced in the MCU-NOMA-IM system with four UAVs. Since the bits carried on the SAP can achieve a lower error probability than those on the EAP, the SAP is still assigned to the farthest UAV. Then, the transmitted signal vector $\mathbf{x}$ is broadcast to four UAVs in time slot t_1 , which is given by

$$\mathbf{x} = \frac{1}{\sqrt{I}} \left[\sqrt{l_1} s_1, \sqrt{l_2} s_2, \sqrt{l_3} a_1, \dots, \sqrt{l_n} a_{(n-k)} \right]^T, \quad (10)$$

where $\mathbf{s} = [s_1, \dots, s_k]$ and $\mathbf{a} = [a_1, \dots, a_{(n-k)}]$ represent the modulated symbol of UAV 1 and UAV 2, respectively. Similar to the case of three UAVs, the received signal vectors in the frequency domain can be written as (2). Likewise, the transmitted signal for UAV 1 can be optimally detected as

$$[\hat{\mathcal{I}}_{u_1}^{t_1}, \hat{\mathbf{s}}_{u_1}^{t_1}, \hat{\mathbf{a}}_{u_1}^{t_1}] = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}, \zeta} \|\mathbf{y}_{u_1}^{t_1} - \mathbf{H}_{s_1} \mathbf{x}\|^2. \quad (11)$$

We note that $\hat{\mathbf{s}}_{u_1}^{t_1}$ is used to recover the information bits of UAV 1, while $\hat{\mathcal{I}}_{u_1}^{t_1}$ and $\hat{\mathbf{a}}_{u_1}^{t_1}$ are the estimates of $\mathcal{I}$ and $\mathbf{a}$ at UAV 1 in time slot t_1 , respectively, which will be constructed as the cooperative vector in the cooperation phase.

2) *Cooperation Phase*: For the MCU-NOMA-IM scheme with four UAVs, the cooperation phase consists of three time slots ($t_2 \sim t_4$). In time slot t_2 , UAV 1 transmits an IM signal vector $\mathbf{f} = [0, \hat{a}_1, \dots, 0, \dots, \hat{a}_{(n-k)}]^T$, constructed by $\hat{\mathcal{I}}_{u_1}^{t_1}$ and $\hat{\mathbf{a}}_{u_1}^{t_1}$, to UAV 2, yielding the received signal vector

$$\mathbf{y}_{u_2}^{t_2} = \mathbf{H}_{u_{12}} \mathbf{f} + \mathbf{n}_{u_{12}}^{t_2}. \quad (12)$$

By receiving the cooperative vector $\mathbf{f}$, the transmitted signals for UAV 2 can be estimated as

$$[\hat{\zeta}_{u_2}^{t_2}, \hat{\mathbf{a}}_{u_2}^{t_2}] = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}, \zeta} \left\{ \|\mathbf{y}_{u_2}^{t_2} - \mathbf{H}_{s_2} \mathbf{x}\|^2 + \|\mathbf{y}_{u_2}^{t_2} - \mathbf{H}_{u_{12}} \mathbf{f}\|^2 \right\}, \quad (13)$$

where $\hat{\mathbf{a}}_{u_2}^{t_2}$ is used to retrieve the information bits of UAV 2. By using $\hat{\zeta}_{u_2}^{t_2}$, a new cooperative signal $\mathbf{g}$ is reconstructed at UAV 2 and then forwarded to UAV 3, which is denoted as

$$\mathbf{g} = \frac{1}{\sqrt{I}} \left[\sqrt{\hat{l}_1}, \sqrt{\hat{l}_2}, \dots, \sqrt{\hat{l}_n} \right]^T. \quad (14)$$

Here, the received signal vector at UAV 3 can be written as

$$\mathbf{y}_{u_3}^{t_3} = \mathbf{H}_{u_{23}} \mathbf{g} + \mathbf{n}_{u_{23}}^{t_3}. \quad (15)$$

By combining (2) and (15), the estimated EAP for UAV 3 can be obtained as

$$[\hat{\mathcal{I}}_{u_3}^{t_3}, \hat{\zeta}_{u_3}^{t_3}] = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}, \zeta} \left\{ \|\mathbf{y}_{u_3}^{t_3} - \mathbf{H}_{s_3} \mathbf{x}\|^2 + \|\mathbf{y}_{u_3}^{t_3} - \mathbf{H}_{u_{23}} \mathbf{g}\|^2 \right\}. \quad (16)$$

In time slot t_4 , UAV 3 transmits an SSK signal $\mathbf{v}$ generated by $\hat{\mathcal{I}}_{u_3}^{t_3}$ to UAV 4, whose expression is similar to (7). The received signal vector at UAV 4 is given by

$$\mathbf{y}_{u_4}^{t_4} = \mathbf{H}_{u_{34}} \mathbf{v} + \mathbf{n}_{u_{34}}^{t_4}. \quad (17)$$

By combining (2) and (17), the estimated SAP for UAV 4 can be obtained as

$$\hat{\mathcal{I}}_{u_4}^{t_4} = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}, \zeta} \left\{ \|\mathbf{y}_{u_4}^{t_4} - \mathbf{H}_{s_4} \mathbf{x}\|^2 + \|\mathbf{y}_{u_4}^{t_4} - \mathbf{H}_{u_{34}} \mathbf{v}\|^2 \right\}, \quad (18)$$

where the information bits of UAV 4 can be recovered by decoding $\hat{\mathcal{I}}_{u_4}^{t_4}$.

III. PERFORMANCE ANALYSIS

In this section, we derive an upper bound for the BER of the MCU-NOMA-IM scheme in the presence of three and four UAVs, assuming perfect channel estimation.

A. Upper Bound for the BER With Three UAVs

1) *BER of UAV 1*: It can be seen from (3) that the estimation errors of the signal $\mathbf{s}$ of UAV 1 can be categorized based on whether the index $\mathcal{I}$ is correctly estimated. Therefore, the BER of UAV 1 satisfies

$$P_{u_1} \leq \frac{1}{2m_3} \sum_{\mathcal{I}, \hat{\mathcal{I}}} \left\{ \Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}}\} \left(P_s N(\mathcal{I}, \hat{\mathcal{I}}) + \frac{1}{2} \bar{N}(\mathcal{I}, \hat{\mathcal{I}}) \right) \right\}, \quad (19)$$

where $\Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}}\}$ denotes the probability of index estimation error; $N(\mathcal{I}, \hat{\mathcal{I}})$ and $\bar{N}(\mathcal{I}, \hat{\mathcal{I}})$ denote the numbers of correctly or incorrectly detected activated subcarriers between $\mathcal{I}$ and $\hat{\mathcal{I}}$; P_s is the average bit error probability of M -ary PSK or QAM symbols, whose closed-form expression has been derived in [45]. It is also assumed that the BER of UAV 1 is 1/2, when the activated subcarriers are incorrectly estimated. An upper bound for $\Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}}\}$ can be derived according to the union bounding technique as [46]

$$\Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}}\} \leq \frac{1}{2^{m_1+m_2}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1}, \quad (20)$$

where $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1}$ is the pairwise error probability (PEP) of erroneously detecting $\mathbf{x}$ as $\hat{\mathbf{x}}$. To obtain $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1}$, we first calculate the conditional PEP as

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | \mathbf{H}_{s_1}\} &= \Pr\left\{ \|\mathbf{y}_{u_1}^{t_1} - \mathbf{H}_{s_1} \mathbf{x}\|^2 > \|\mathbf{y}_{u_1}^{t_1} - \mathbf{H}_{s_1} \hat{\mathbf{x}}\|^2 \right\} \\ &= Q \left(\sqrt{\frac{\|\mathbf{H}_{s_1}(\mathbf{x} - \hat{\mathbf{x}})\|^2}{2N_0}} \right). \end{aligned} \quad (21)$$

For ease of analysis, we define

$$\lambda = \|\mathbf{H}_{s_1}(\mathbf{x} - \hat{\mathbf{x}})\|^2 = \sum_{t=1}^n (x(t) - \hat{x}(t))^2 |h_{s_1}(t)|^2, \quad (22)$$

where $h_{s_1}(t)$ is the t -th element of $\mathbf{h}_{s_1}$. Since a Nakagami- m fading channel is considered, $|h_{s_1}(t)|^2$ obeys the gamma distribution, i.e.,

$$|h_{s_1}(t)|^2 \sim \mathcal{G} \left(m_{s_1}, \frac{\Omega_{s_1}}{m_{s_1}} \right). \quad (23)$$

At this point, λ does not follow any specific distribution because of the term $(x(t) - \hat{x}(t))^2$. However, to derive the unconditional PEP from (21), it is essential to use the probability density function of λ . Hence, for ease of analysis, we only discuss the case with $m_\Delta = 1$ for $\Delta \in \{s_i, u_{ij}\}$ in this section. Under this assumption, the channel coefficient $h_\Delta(t)$ can be rewritten as a complex Gaussian distributed random variable with zero mean and variance Ω_Δ . Applying the approximation $Q(x) \cong \frac{1}{12}e^{-\frac{x^2}{2}} + \frac{1}{4}e^{-\frac{2x^2}{3}}$ derived in [47] for the Q -function yields the unconditional PEP

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1} &= E_{\mathbf{H}_{s_1}} \left\{ \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | \mathbf{H}_{s_1}\}_{u_1} \right\} \\ &= \frac{1/12}{(\det(\mathbf{I}_n + q_1 \mathbf{K} \mathbf{A}))} + \frac{1/4}{(\det(\mathbf{I}_n + q_2 \mathbf{K} \mathbf{A}))}, \end{aligned} \quad (24)$$

where $q_1 = \frac{1}{2N_0}$ and $q_2 = \frac{2}{3N_0}$; $\mathbf{K} = E\{\mathbf{H}_{s_1} \mathbf{H}_{s_1}^H\}$ is the covariance matrix of $\mathbf{H}_{s_1}$; $\mathbf{A} = \text{diag}\{\mathbf{x} - \hat{\mathbf{x}}\}^H \text{diag}\{\mathbf{x} - \hat{\mathbf{x}}\}$. Substituting (20) and (24) into (19) yields the closed-form expression for the BER upper bound of UAV 1.

2) *BER of UAV 2*: As for UAV 2, the error events can be classified into three categories:

- UAV 1 incorrectly estimates the SAP $\mathcal{I}$;
- UAV 1 incorrectly estimates the symbols $\mathbf{a}$ while the SAP $\mathcal{I}$ is correctly estimated;
- UAV 2 incorrectly estimates the symbols $\mathbf{a}$ while the SAP $\mathcal{I}$ and the symbols $\mathbf{a}$ are correctly estimated at UAV 1.

Therefore, the BER of UAV 2 can be jointly written as

$$P_{u_2} = \frac{1}{2}P_{\mathcal{I}}^{u_1} + \frac{1}{2}P_{\mathcal{I},\mathbf{a}}^{u_1} + (1 - P_{\mathcal{I}}^{u_1} - P_{\mathcal{I},\mathbf{a}}^{u_1})P_{\mathbf{a}}^{u_2}, \quad (25)$$

where $P_{\mathcal{I}}^{u_1}$ is the error probability of the SAP estimation at UAV 1; $P_{\mathcal{I},\mathbf{a}}^{u_1}$ is the error probability of symbol estimation for UAV 2 at UAV 1 when the SAP $\mathcal{I}$ is correctly estimated at UAV 1; $P_{\mathbf{a}}^{u_2}$ is the BER of UAV 2 when the SAP $\mathcal{I}$ and symbols $\mathbf{a}$ are correctly estimated at UAV 1. Note that the signal detection of UAV 2 largely relies on the cooperation link due to the short transmission distance between UAV 1 and UAV 2. Therefore, the BER of UAV 2 can be approximated to be 1/2, when the SAP $\mathcal{I}$ and/or modulated symbols $\mathbf{a}$ are incorrectly estimated at UAV 1. Upper bounds for $P_{\mathcal{I}}^{u_1}$ and $P_{\mathcal{I},\mathbf{a}}^{u_1}$ can be derived as [46]

$$P_{\mathcal{I}}^{u_1} \leq \frac{1}{2^m} \sum_{\substack{\mathcal{I}, \hat{\mathcal{I}} \\ \mathcal{I} \neq \hat{\mathcal{I}}}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1}, \quad (26)$$

and

$$P_{\mathcal{I},\mathbf{a}}^{u_1} \leq \frac{1}{2^m} \sum_{\substack{\mathcal{I}, \hat{\mathcal{I}} \\ \mathcal{I} = \hat{\mathcal{I}}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1}, \quad (27)$$

respectively, where $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}\}_{u_1}$ is given in (24). Besides, an upper bound for $P_{\mathbf{a}}^{u_2}$ can be derived as

$$\begin{aligned} P_{\mathbf{a}}^{u_2} &\leq \frac{1}{2^{m_3}} \sum_{\mathcal{I}, \hat{\mathcal{I}}} \left\{ \Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}} | E_0\}_{u_2} \right. \\ &\quad \times \left. \left(P_s N(\mathcal{I}, \hat{\mathcal{I}}) + \frac{1}{2} \bar{N}(\mathcal{I}, \hat{\mathcal{I}}) \right) \right\}, \end{aligned} \quad (28)$$

where E_0 denotes the event that the SAP $\mathcal{I}$ and symbols $\mathbf{a}$ for UAV 2 are correctly estimated at UAV 1; $\Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}} | E_0\}_{u_2}$ denotes the error probability of the SAP estimation at UAV 2 under the condition of E_0 , which can be expressed as

$$\Pr\{\mathcal{I} \rightarrow \hat{\mathcal{I}} | E_0\}_{u_2} \leq \frac{\sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}} | E_0\}_{u_2}}{2^{m_1+m_2}}, \quad (29)$$

where $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}} | E_0\}_{u_2}$ is the PEP of erroneously detecting $\{\mathbf{x}, \mathbf{f}\}$ as $\{\hat{\mathbf{x}}, \hat{\mathbf{f}}\}$, whose conditional PEP can be expressed as

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}} | E_0, \mathbf{H}_{s_2}, \mathbf{H}_{u_{12}}\}_{u_2} &= \Pr\left\{ \|\mathbf{y}_{u_2}^{t_1} - \mathbf{H}_{s_2} \mathbf{x}\|^2 + \|\mathbf{y}_{u_2}^{t_2} - \mathbf{H}_{u_{12}} \mathbf{f}\|^2 \right. \\ &> \|\mathbf{y}_{u_2}^{t_1} - \mathbf{H}_{s_2} \hat{\mathbf{x}}\|^2 + \|\mathbf{y}_{u_2}^{t_2} - \mathbf{H}_{u_{12}} \hat{\mathbf{f}}\|^2 \left. \right\} \\ &= Q\left(\sqrt{\frac{\tau_1}{2N_0}}\right), \end{aligned} \quad (30)$$

where $\tau_1 = \|\mathbf{H}_{s_2}(\mathbf{x} - \hat{\mathbf{x}})\|^2 + \|\mathbf{H}_{u_{12}}(\mathbf{f} - \hat{\mathbf{f}})\|^2$. Note that when $m_\Delta = 1$ with $\Delta \in \{s_i, u_{ij}\}$, τ_1 is the sum of two independently central chi-square distributed random variables, whose moment generating function (MGF) can be expressed as follows [48]:

$$\mathcal{M}_{\tau_1}(\mu, t) = \frac{1}{(1 - 2\mu\lambda_1^2(t))(1 - 2\mu\lambda_2^2(t))}, \quad (31)$$

where $\lambda_1^2(t) = \Omega_{s_2}|x(t) - \hat{x}(t)|^2$ and $\lambda_2^2(t) = \Omega_{u_{12}}|f(t) - \hat{f}(t)|^2$ with $1 \leq t \leq n$. Applying the MGF and the approximation for the Q -function, the unconditional PEP can be derived in closed form as

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}} | E_0\}_{u_2} &\approx \frac{1}{12} \prod_{t=1}^n \mathcal{M}_{\tau_1}(-q_1, t) + \frac{1}{4} \prod_{t=1}^n \mathcal{M}_{\tau_1}(-q_2, t). \end{aligned} \quad (32)$$

Finally, the BER upper bounds of UAV 2 can be obtained by substituting (26), (28), (29), and (32) into (25).

3) *BER of UAV 3*: Similarly, the error events of UAV 3 can be classified into five categories:

- UAV 1 incorrectly estimates the SAP $\mathcal{I}$;
- UAV 2 incorrectly estimates the SAP $\mathcal{I}$ when UAV 1 correctly estimates the SAP $\mathcal{I}$ and the modulated symbol vector $\mathbf{a}$;
- UAV 2 incorrectly estimates the SAP $\mathcal{I}$ when UAV 1 correctly estimates the SAP $\mathcal{I}$ and incorrectly estimates the modulated symbol vector $\mathbf{a}$;

- UAV 3 incorrectly estimates the SAP $\mathcal{I}$ when the SAP $\mathcal{I}$ is correctly estimated at both UAV 1 and UAV 2, and the modulated symbol vector $\mathbf{a}$ is also correctly estimated at UAV 1;
- UAV 3 incorrectly estimates the SAP $\mathcal{I}$ when the SAP is correctly estimated at both UAV 1 and UAV 2, yet the modulated symbol vector $\mathbf{a}$ is incorrectly estimated at UAV 1.

For ease of presentation, we first outline three different conditions for the derivation of BER:

- Condition E_1 denotes that the SAP $\mathcal{I}$ and the modulated symbol vector $\mathbf{a}$ are correctly and incorrectly estimated at UAV 1, respectively;
- Condition E_2 denotes that the SAP $\mathcal{I}$ is correctly estimated at UAV 2;
- Condition E_3 denotes that the SAP $\mathcal{I}$ and the modulated symbol vector $\mathbf{a}$ are correctly estimated at UAV 1.

By considering the five error events for UAV 3, the BER can be expressed as

$$P_{u_3} = \frac{1}{2}P_{\tilde{\mathcal{I}}}^{u_1} + \frac{1}{2}P_{\mathcal{I},\tilde{\mathbf{a}}}^{u_1}P_{\tilde{\mathcal{I}},E_1}^{u_2} + \frac{1}{m_3}P_{\mathcal{I},\tilde{\mathbf{a}}}^{u_1}(1 - P_{\tilde{\mathcal{I}},E_1}^{u_2})P_{\tilde{\mathcal{I}},E_2}^{u_3} + \frac{1}{2}(1 - P_{\tilde{\mathcal{I}}}^{u_1} - P_{\mathcal{I},\tilde{\mathbf{a}}}^{u_1})P_{\tilde{\mathcal{I}},E_3}^{u_2} + \frac{1}{m_3}(1 - P_{\tilde{\mathcal{I}}}^{u_1} - P_{\mathcal{I},\tilde{\mathbf{a}}}^{u_1})(1 - P_{\tilde{\mathcal{I}},E_3}^{u_2})P_{\tilde{\mathcal{I}},E_2}^{u_3}, \quad (33)$$

where $P_{\tilde{\mathcal{I}},E_1}^{u_2}$ (or $P_{\tilde{\mathcal{I}},E_3}^{u_2}$) is the error probability of the SAP estimation at UAV 2 under the condition E_1 (or E_3), and $P_{\tilde{\mathcal{I}},E_2}^{u_3}$ is the error probability of the SAP estimation at UAV 3 under the condition E_2 . Both $P_{\tilde{\mathcal{I}}}^{u_1}$ and $P_{\mathcal{I},\tilde{\mathbf{a}}}^{u_1}$ have been derived in (26). Due to the erroneous estimation of the transmitted symbol vector $\mathbf{a}$ at UAV 1, UAV 2 only needs to consider the cooperative link for detection, given that UAV 1 is closer to UAV 2. An upper bound for $P_{\tilde{\mathcal{I}},E_1}^{u_2}$ can be derived as

$$P_{\tilde{\mathcal{I}},E_1}^{u_2} \leq \frac{1}{2^{m_2+m_3}} \sum_{\mathcal{I},\tilde{\mathcal{I}}} \sum_{\mathbf{a},\tilde{\mathbf{a}}} \Pr\{\mathbf{f} \rightarrow \hat{\mathbf{f}}|E_1\}_{u_2}, \quad (34)$$

where $\Pr\{\mathbf{f} \rightarrow \hat{\mathbf{f}}|E_1\}_{u_2}$ is the PEP of erroneously detecting $\mathbf{f}$ as $\hat{\mathbf{f}}$ at UAV 2 under the condition E_1 , which is given by

$$\Pr\{\mathbf{f} \rightarrow \hat{\mathbf{f}}|E_1\}_{u_2} = E_{\mathbf{H}_{u_{12}}} \left\{ Q \left(\sqrt{\frac{\tau_2}{2N_0}} \right) \right\} \approx \frac{1/12}{(\det(\mathbf{I}_n + q_1 \mathbf{K}_2 \mathbf{B}))} + \frac{1/4}{(\det(\mathbf{I}_n + q_2 \mathbf{K}_2 \mathbf{B}))}, \quad (35)$$

where $\tau_2 = \|\mathbf{H}_{u_{12}}(\mathbf{f} - \hat{\mathbf{f}})\|^2$; $\mathbf{K}_2 = E\{\mathbf{H}_{u_{12}}\mathbf{H}_{u_{12}}^H\}$ is the covariance matrix of $\mathbf{H}_{u_{12}}$; $\mathbf{B} = \text{diag}\{\mathbf{f} - \hat{\mathbf{f}}\}^H \text{diag}\{\mathbf{f} - \hat{\mathbf{f}}\}$. An upper bound for $P_{\tilde{\mathcal{I}},E_2}^{u_3}$ can be derived as

$$P_{\tilde{\mathcal{I}},E_2}^{u_3} \leq \frac{1}{2^m} \sum_{\mathcal{I},\tilde{\mathcal{I}}} \sum_{\mathbf{s},\tilde{\mathbf{s}}} \sum_{\mathbf{a},\tilde{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{q} \rightarrow \hat{\mathbf{q}}|E_2\}_{u_3}, \quad (36)$$

where $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{q} \rightarrow \hat{\mathbf{q}}|E_2\}_{u_3}$ denotes the PEP of erroneously detecting $\{\mathbf{x}, \mathbf{q}\}$ as $\{\hat{\mathbf{x}}, \hat{\mathbf{q}}\}$ at UAV 3 under

the condition E_2 , which can be expressed as

$$\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{q} \rightarrow \hat{\mathbf{q}}|E_2\}_{u_3} = E_{\mathbf{H}_{u_{23}}, \mathbf{H}_{s_3}} \left\{ Q \left(\sqrt{\frac{\tau_3}{2N_0}} \right) \right\} \approx \frac{1}{12} \prod_{t=1}^n \mathcal{M}_{\tau_3}(-q_1, t) + \frac{1}{4} \prod_{t=1}^n \mathcal{M}_{\tau_3}(-q_2, t), \quad (37)$$

where $\tau_3 = \|\mathbf{H}_{s_3}(\mathbf{x} - \hat{\mathbf{x}})\|^2 + \|\mathbf{H}_{u_{23}}(\mathbf{q} - \hat{\mathbf{q}})\|^2$; $\mathcal{M}_{\tau_3}(\mu, t) = \frac{1}{(1-2\mu\lambda_3^2(t))(1-2\mu\lambda_4^2(t))}$, with $\lambda_3^2(t) = \Omega_{s_3}(x(t) - \hat{x}(t))^2$ and $\lambda_4^2(t) = \Omega_{u_{23}}(q(t) - \hat{q}(t))^2$, for $1 \leq t \leq n$. An upper bound for $P_{\tilde{\mathcal{I}},E_3}^{u_2}$ can be derived as

$$P_{\tilde{\mathcal{I}},E_3}^{u_2} \leq \frac{1}{2^m} \sum_{\mathcal{I},\tilde{\mathcal{I}}} \sum_{\mathbf{s},\tilde{\mathbf{s}}} \sum_{\mathbf{a},\tilde{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}}|E_3\}_{u_2}, \quad (38)$$

where $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}}|E_3\}_{u_2}$ denotes the PEP when erroneously detecting $\{\mathbf{x}, \mathbf{f}\}$ as $\{\hat{\mathbf{x}}, \hat{\mathbf{f}}\}$ at UAV 2, which can be expressed as

$$\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{f} \rightarrow \hat{\mathbf{f}}|E_3\}_{u_2} = E_{\mathbf{H}_{u_{12}}, \mathbf{H}_{s_2}} \left\{ Q \left(\sqrt{\frac{\tau_4}{2N_0}} \right) \right\} \approx \frac{1}{12} \prod_{t=1}^n \mathcal{M}_{\tau_4}(-q_1, t) + \frac{1}{4} \prod_{t=1}^n \mathcal{M}_{\tau_4}(-q_2, t), \quad (39)$$

where $\tau_4 = \|\mathbf{H}_{s_2}(\mathbf{x} - \hat{\mathbf{x}})\|^2 + \|\mathbf{H}_{u_{12}}(\mathbf{f} - \hat{\mathbf{f}})\|^2$; $\mathcal{M}_{\tau_4}(\mu, t) = \frac{1}{(1-2\mu\lambda_5^2(t))(1-2\mu\lambda_6^2(t))}$, with $\lambda_5^2(t) = \Omega_{s_2}(x(t) - \hat{x}(t))^2$ and $\lambda_6^2(t) = \Omega_{u_{12}}(f(t) - \hat{f}(t))^2$, for $1 \leq t \leq n$. By substituting (26), (34), (36), and (38) into (33), the BER of UAV 3 can finally be obtained in closed form.

B. Upper Bound for the BER With Four UAVs

Since the intended information for UAV 1 and UAV 2 is the same in both the three-UAV and four-UAV cases, their performance analyses are identical in these scenarios, as provided in (19) and (25). For brevity, this subsection focuses on the performance analysis of UAV 3 and UAV 4.

1) *BER of UAV 3:* Since the cooperation between UAV 1 and UAV 2 is irrelevant to the cooperation between UAV 2 and UAV 3, the error event on the estimation of the EAP at UAV 3 is only determined by UAV 2 and UAV 3. Therefore, the error events of UAV 3 can only be classified into two categories: a) UAV 2 incorrectly estimates the EAP; b) UAV 3 incorrectly estimates the EAP while UAV 2 correctly estimated the EAP. Therefore, the BER of UAV 3 can be expressed as

$$P_{u_3}^{4u} = \frac{1}{2}F_{\tilde{\zeta}}^{u_2} + \frac{1}{m_3}(1 - F_{\tilde{\zeta}}^{u_2})F_{\tilde{\zeta}}^{u_3}, \quad (40)$$

where $F_{\tilde{\zeta}}^{u_2}$ represents the error probability of the EAP estimation at UAV 2, and $F_{\tilde{\zeta}}^{u_3}$ indicates the error probability of the EAP estimation at UAV 3 when the EAP is correctly estimated

at UAV 2. An upper bound for $F_{\hat{\zeta}}^{u_2}$ can be derived as

$$F_{\hat{\zeta}}^{u_2} \leq \frac{1}{2^m} \sum_{\substack{\zeta, \hat{\zeta} \\ \zeta \neq \hat{\zeta}}} \sum_{\mathcal{I}, \hat{\mathcal{I}}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | E_4\}_{u_2}, \quad (41)$$

where E_4 refers to the event that the EAP is incorrectly estimated at UAV 2; $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | E_4\}_{u_2}$ denotes the PEP of incorrectly detecting $\mathbf{x}$ as $\hat{\mathbf{x}}$ at UAV 2 under the condition E_4 , which can be expressed as

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | E_4\}_{u_2} &= E_{\mathbf{H}_{s_2}} \left\{ Q \left(\sqrt{\frac{\tau_5}{2N_0}} \right) \right\} \\ &\approx \frac{1/12}{(\det(\mathbf{I}_n + q_1 \mathbf{K}_3 \mathbf{C}))} + \frac{1/4}{(\det(\mathbf{I}_n + q_2 \mathbf{K}_3 \mathbf{C}))}, \end{aligned} \quad (42)$$

where $\tau_5 = \|\mathbf{H}_{s_2}(\mathbf{x} - \hat{\mathbf{x}})\|^2$; $\mathbf{C} = \text{diag}\{\mathbf{x} - \hat{\mathbf{x}}\}^H \text{diag}\{\mathbf{x} - \hat{\mathbf{x}}\}$; $\mathbf{K}_3 = E\{\mathbf{H}_{s_2} \mathbf{H}_{s_2}^H\}$ is the covariance matrix of $\mathbf{H}_{s_2}$. Note that the cooperative link between UAV 1 and UAV 2 is not considered for the EAP estimation due to the independence between the estimations of the EAP and modulation symbols. Here, an upper bound for $F_{\hat{\zeta}}^{u_3}$ can be derived as

$$F_{\hat{\zeta}}^{u_3} \leq \frac{1}{2^m} \sum_{\substack{\zeta, \hat{\zeta} \\ \zeta \neq \hat{\zeta}}} \sum_{\mathcal{I}, \hat{\mathcal{I}}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{g} \rightarrow \hat{\mathbf{g}} | E_5\}_{u_3}, \quad (43)$$

where E_5 refers to the condition that the EAP is correctly estimated at UAV 2; $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{g} \rightarrow \hat{\mathbf{g}} | E_5\}_{u_3}$ denotes the PEP of incorrectly detecting $\{\mathbf{x}, \mathbf{g}\}$ as $\{\hat{\mathbf{x}}, \hat{\mathbf{g}}\}$ at UAV 3 under the condition E_5 , which is given by

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{g} \rightarrow \hat{\mathbf{g}} | E_5\}_{u_3} &= E_{\mathbf{H}_{u_{23}}, \mathbf{H}_{s_3}} \left\{ Q \left(\sqrt{\frac{\tau_6}{2N_0}} \right) \right\} \\ &\approx \frac{1}{12} \prod_{t=1}^n \mathcal{M}_{\tau_6}(-q_1, t) + \frac{1}{4} \prod_{t=1}^n \mathcal{M}_{\tau_6}(-q_2, t), \end{aligned} \quad (44)$$

where $\tau_6 = \|\mathbf{H}_{s_3}(\mathbf{x} - \hat{\mathbf{x}})\|^2 + \|\mathbf{H}_{u_{23}}(\mathbf{g} - \hat{\mathbf{g}})\|^2$; $\mathcal{M}_{\tau_6}(\mu, t) = \frac{1}{(1-2\mu\lambda_7^2(t))(1-2\mu\lambda_8^2(t))}$ with $\lambda_7^2(t) = \Omega_{s_3}(x(t) - \hat{x}(t))^2$ and $\lambda_8^2(t) = \Omega_{u_{23}}(g(t) - \hat{g}(t))^2$ for $1 \leq t \leq n$. By substituting (41) and (43) into (40), the BER of UAV 3 in the four-UAV case can be finally obtained in closed form.

2) *BER of UAV 4*: Similarly, the error occurrence on the index estimation at UAV 4 is only dependent on UAV 3 and UAV 4. Therefore, the estimation errors at UAV 4 can still be classified into two categories: a) UAV 3 incorrectly estimates the SAP $\mathcal{I}$; b) UAV 4 incorrectly estimates the SAP $\mathcal{I}$, while UAV 3 correctly estimated the SAP $\mathcal{I}$. Therefore, the BER of UAV 4 can be written as

$$P_{u_4}^{4u} = \frac{1}{2} F_{\hat{\mathcal{I}}}^{u_3} + \frac{1}{m_4} (1 - F_{\hat{\mathcal{I}}}^{u_3}) F_{\hat{\mathcal{I}}}^{u_4}, \quad (45)$$

where $F_{\hat{\mathcal{I}}}^{u_3}$ is the error probability of index estimation at UAV 3, and $F_{\hat{\mathcal{I}}}^{u_4}$ represents the error probability of index estimation at UAV 4 when the SAP is correctly estimated at

UAV 3. As a result, an upper bound for $F_{\hat{\mathcal{I}}}^{u_3}$ can be derived as

$$F_{\hat{\mathcal{I}}}^{u_3} \leq \frac{1}{2^m} \sum_{\substack{\mathcal{I}, \hat{\mathcal{I}} \\ \mathcal{I} \neq \hat{\mathcal{I}}}} \sum_{\zeta, \hat{\zeta}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | E_6\}_{u_3}, \quad (46)$$

where E_6 refers to the condition that the SAP is incorrectly estimated at UAV 3; $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | E_6\}_{u_3}$ denotes the PEP of incorrectly detecting $\mathbf{x}$ as $\hat{\mathbf{x}}$ at UAV 3 under the condition E_6 , which can be expressed as

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}} | E_6\}_{u_3} &= E_{\mathbf{H}_{s_3}} \left\{ Q \left(\sqrt{\frac{\tau_7}{2N_0}} \right) \right\} \\ &\approx \frac{1/12}{(\det(\mathbf{I}_n + q_1 \mathbf{K}_4 \mathbf{C}))} + \frac{1/4}{(\det(\mathbf{I}_n + q_2 \mathbf{K}_4 \mathbf{C}))}, \end{aligned} \quad (47)$$

where $\tau_7 = \|\mathbf{H}_{s_3}(\mathbf{x} - \hat{\mathbf{x}})\|^2$; $\mathbf{C} = \text{diag}\{\mathbf{x} - \hat{\mathbf{x}}\}^H \text{diag}\{\mathbf{x} - \hat{\mathbf{x}}\}$; $\mathbf{K}_4 = E\{\mathbf{H}_{s_3} \mathbf{H}_{s_3}^H\}$ is the covariance matrix of $\mathbf{H}_{s_3}$. Note that the cooperative link between UAV 2 and UAV 3 will not affect the SAP estimation due to the independence between the estimations of the EAP and SAP. Then, an upper bound for $F_{\hat{\mathcal{I}}}^{u_4}$ can be deduced as

$$F_{\hat{\mathcal{I}}}^{u_4} \leq \frac{1}{2^m} \sum_{\substack{\mathcal{I}, \hat{\mathcal{I}} \\ \mathcal{I} \neq \hat{\mathcal{I}}}} \sum_{\zeta, \hat{\zeta}} \sum_{\mathbf{s}, \hat{\mathbf{s}}} \sum_{\mathbf{a}, \hat{\mathbf{a}}} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{v} \rightarrow \hat{\mathbf{v}} | E_7\}_{u_4}, \quad (48)$$

where E_7 refers to the condition that the SAP is correctly estimated at UAV 3; $\Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{v} \rightarrow \hat{\mathbf{v}} | E_7\}_{u_4}$ denotes the PEP of detecting $\{\mathbf{x}, \mathbf{v}\}$ as $\{\hat{\mathbf{x}}, \hat{\mathbf{v}}\}$ by mistake at UAV 4 under the condition E_7 , which can be written as

$$\begin{aligned} \Pr\{\mathbf{x} \rightarrow \hat{\mathbf{x}}, \mathbf{v} \rightarrow \hat{\mathbf{v}} | E_7\}_{u_4} &= E_{\mathbf{H}_{u_{34}}, \mathbf{H}_{s_4}} \left\{ Q \left(\sqrt{\frac{\tau_8}{2N_0}} \right) \right\} \\ &\approx \frac{1}{12} \prod_{t=1}^n \mathcal{M}_{\tau_8}(-q_1, t) + \frac{1}{4} \prod_{t=1}^n \mathcal{M}_{\tau_8}(-q_2, t), \end{aligned} \quad (49)$$

where $\tau_8 = \|\mathbf{H}_{s_4}(\mathbf{x} - \hat{\mathbf{x}})\|^2 + \|\mathbf{H}_{u_{34}}(\mathbf{v} - \hat{\mathbf{v}})\|^2$; $\mathcal{M}_{\tau_8}(\mu, t) = \frac{1}{(1-2\mu\lambda_9^2(t))(1-2\mu\lambda_{10}^2(t))}$ with $\lambda_9^2(t) = \Omega_{s_4}(x(t) - \hat{x}(t))^2$ and $\lambda_{10}^2(t) = \Omega_{u_{34}}(v(t) - \hat{v}(t))^2$ for $1 \leq t \leq n$. The BER of UAV 4 can be derived in closed form by substituting (46) and (48) into (45).

One can easily observe from the analysis of the system models with three and four UAVs that the BER is dominated by the direct link from the GS and the cooperation link established with near UAVs, excluding the nearest one. This cooperation link may in turn be influenced by another near UAV, as shown in these scenarios. Therefore, when more than four UAVs are present, the theoretical analysis of the cooperation links for each UAV remains similar to that of the considered cases with three and four UAVs. Hence, we take these cases as examples to illustrate the theoretical analysis.

IV. ENHANCED MCU-NOMA-IM SCHEME

When adopting MCU-NOMA-IM in air-ground networks with K UAVs, the whole transmission process requires K time slots to fully exploit the advantage of cross-UAV cooperation,

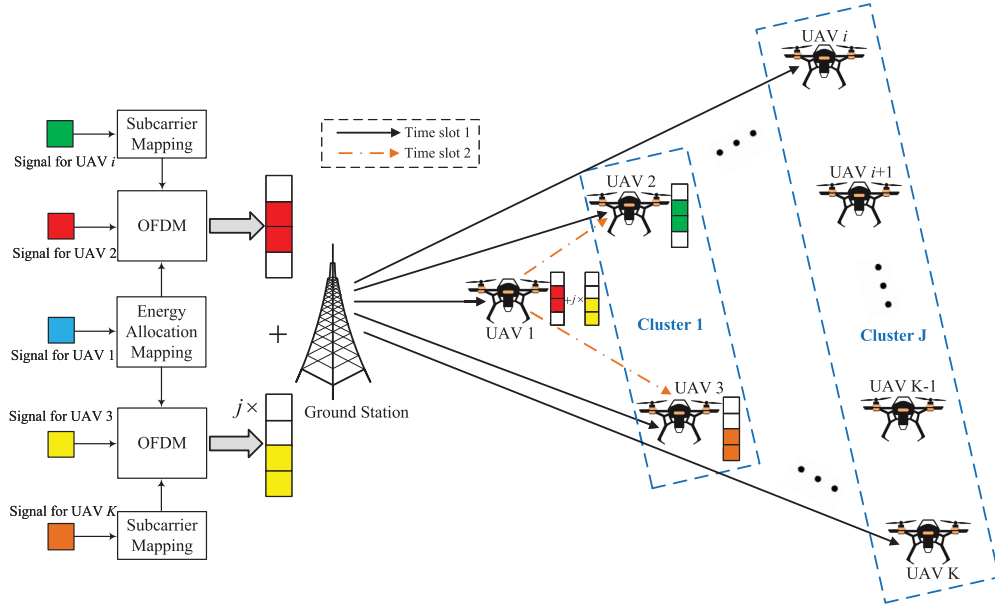


Fig. 2. System model of the considered MCCU-NOMA-IM system with K UAVs.

which might lead to a high latency for the UAVs close to the cell edge. This is especially true when the number of UAVs is large. However, most UAV-aided communication scenarios are latency-critical, such as emergency communications and intelligent traffic communications, which limits the feasibility of MCU-NOMA-IM. Therefore, in this section, we propose an enhanced scheme, termed MCCU-NOMA-IM⁴ as shown in Fig. 2, to considerably reduce the number of time slots by clustering closely located UAVs. Assume that in MCCU-NOMA-IM, UAV 1 is the nearest to the GS for cooperation purposes, and that the other $(K - 1)$ UAVs are grouped into J clusters, denoted as $\{C_1, \dots, C_J\}$, where C_1 and C_J denote the nearest and farthest clusters, respectively, so that the UAVs within each cluster are located at a similar distance from the GS. Accordingly, we have $\Omega_{s_1} > \Omega_{s_{i_1}} > \dots > \Omega_{s_{i_J}}$, where s_{i_ϵ} denotes the UAV i_ϵ belonging to cluster C_ϵ with $i_\epsilon \in \{2, \dots, K\}$ and $\epsilon \in \{1, 2, \dots, J\}$. Note that in MCU-NOMA-IM, each UAV can cooperate with only one other UAV within a single time slot. In contrast, MCCU-NOMA-IM allows each UAV to cooperate with multiple UAVs within one time slot, significantly reducing the end-to-end transmission latency. Therefore, we can see that the MCCU-NOMA-IM scheme requires $J + 1$ time slots ($t_1 \sim t_{J+1}$) to complete a single transmission, where t_1 denotes the broadcast phase and $t_2 \sim t_{J+1}$ denote the cooperative phase. For ease of presentation, we consider the setups with $K = 3$ and $K = 5$ in air-ground networks using MCCU-NOMA-IM.⁵

⁴This section primarily focuses on the system design rather than on the theoretical performance of MCCU-NOMA-IM. This is because the theoretical analyses for both MCCU-NOMA-IM and MCU-NOMA-IM are identical, owing to the same signal transmission and reception architectures, as shown in Figs. 1-2.

⁵MCU/MCCU-NOMA-IM can be readily extended to scenarios with multiple UAVs. When the number of UAVs or clusters exceeds the number of independent signal dimensions, the power allocation strategy is employed to accommodate the excess of UAVs. Even if the excess of UAVs is affected by IUI, the proposed scheme still achieves a moderate performance gain over NOMA as the overlap between signals in the power domain is negligible due to the presence of multiple signal dimensions.

A. MCCU-NOMA-IM With $K = 3$ and $J = 1$

In MCCU-NOMA-IM with $K = 3$ and $J = 1$, UAV 2 and UAV 3 are grouped into cluster C_1 , and UAV 1 aims to simultaneously cooperate with UAV 2 and UAV 3 in time slot t_2 . The transmission phases are detailed below.

1) *Broadcast Phase*: During the broadcast phase, the transmission process of MCCU-NOMA-IM is the same as that of MCU-NOMA-IM, where a total of $m = m_1 + m_2 + m_3$ bits are transmitted to the three UAVs within time slot t_1 . Recall that the modulated symbol vectors $\mathbf{s} = [s_1, \dots, s_k]$ for UAV 1 and $\mathbf{a} = [a_1, \dots, a_{(n-k)}]$ for UAV 2 are determined by the m_1 and m_2 bits, respectively, and the SAP $\mathcal{I} = \{i_1, \dots, i_k\}$ for UAV 3 is determined by the m_3 bits. In time slot t_1 , the transmitted signal vector $\mathbf{x}$, the received signal $\mathbf{y}_{u_i}^{t_1}$ (where $i \in 1, \dots, K$), and the estimates of $\mathcal{I}$, $\mathbf{s}$, and $\mathbf{a}$ at UAV 1 can be described by (1), (2), and (3), respectively.

2) *Cooperation Phase*: For $K = 3$, the cooperation phase of MCU-NOMA-IM requires two time slots, whereas MCCU-NOMA-IM needs only one time slot for the same phase. This indicates that UAV 1 can transmit the cooperative vector to both UAV 2 and UAV 3 simultaneously. Accordingly, in time slot t_2 , UAV 1 first constructs a new IM signal vector $\mathbf{f}$ using $\hat{\mathcal{I}}_{u_1}^{t_1}$ and $\hat{\mathbf{a}}_{u_1}^{t_1}$, denoted as $\mathbf{f} = [0, \hat{a}_1, \dots, 0, \dots, \hat{a}_{(n-k)}]^T$, and then forwards $\mathbf{f}$ to UAV 2 and UAV 3, yielding

$$\mathbf{y}_{u_i}^{t_2} = \mathbf{H}_{u_{1i}} \mathbf{f} + \mathbf{n}_{u_{1i}}^{t_2}, \quad i = 2, 3. \quad (50)$$

The signal detection for UAV 2 is the same as that in MCU-NOMA-IM, which can be referred to as (6). However, the signal detection for UAV 3 needs to be updated as

$$\hat{\mathcal{I}}_{u_3}^{t_2} = \arg \min_{\mathcal{I}, \mathbf{s}, \mathbf{a}} \{\|\mathbf{y}_{u_3}^{t_1} - \mathbf{H}_{u_3} \mathbf{x}\|^2 + \|\mathbf{y}_{u_3}^{t_2} - \mathbf{H}_{u_{13}} \mathbf{f}\|^2\}, \quad (51)$$

since the cooperation vector in MCCU-NOMA-IM comes from UAV 1 rather than UAV 2. Here, the information bits for UAV 3 can be obtained by decoding $\hat{\mathcal{I}}_{u_3}^{t_2}$.

B. MCCU-NOMA-IM With $K = 5$ and $J = 2$

To achieve interference-free detection, the signal construction for five UAVs in the MCCU-NOMA-IM system needs to be carefully designed by utilizing multiple independent domains. Therefore, in the considered scenario, we resort to the orthogonality between the in-phase and quadrature domains to allocate the data. In MCCU-NOMA-IM with $K = 5$, UAV 2 and UAV 3 are grouped into cluster C_1 , while UAV 4 and UAV 5 are grouped into cluster C_2 . Again, we have the broadcast phase and the cooperation phase, as detailed below.

1) *Broadcast Phase*: During time slot t_1 , the GS broadcasts an IM vector, derived from a total of p bits, to all UAVs using n subcarriers. These p bits are divided into three parts: the composition part of p^C bits, the in-phase part of p^I bits, and the quadrature part of p^Q bits. In detail, we have the following:

- For the composition part, $p^C = \lfloor \log_2 C(I-1, \alpha-1) \rfloor$ bits for UAV 1 are used to determine the EAP $\zeta = \{\zeta_1, \dots, \zeta_\alpha\}$, where $\{\zeta_\alpha\}_{\alpha=1}^\alpha \in \{1, 2, \dots, I-1\}$ with $\alpha = k^I + k^Q$ and I represents the energy activation factor. Here, k^I and k^Q denote the numbers of active subcarriers for the in-phase and quadrature parts, respectively.
- For the in-phase part, p^I bits can be split into two parts: $p_1^I = k^I \log_2(M^I)$ bits for UAV 2 and $p_2^I = \lfloor \log_2 C(n, k^I) \rfloor$ bits for UAV 4. Specifically, p_1^I bits are used to determine the symbol vector $\mathbf{s}^I = [s_1^I, \dots, s_{k^I}^I]$, where the symbol s_ℓ^I with $\ell = \{1, \dots, k^I\}$ is generated from the constellation set $\mathcal{S}_I$ of the M^I -ary pulse amplitude modulation (PAM). p_2^I bits are used to select the SAP $\mathcal{I}^I = \{i_1^I, \dots, i_{k^I}^I\}$ with $\{i_\tau^I\}_{\tau=1}^{k^I} \in \{1, 2, \dots, n\}$ for conveying the symbol vector $\mathbf{s}^I$ on the k^I active subcarriers.
- For the quadrature part, p^Q bits can also be split as follows: $p_1^Q = k^Q \log_2(M^Q)$ bits for UAV 3 and $p_2^Q = \lfloor \log_2 C(n, k^Q) \rfloor$ bits for UAV 5. Similarly, p_1^Q bits are used to determine the symbol vector $\mathbf{s}^Q = [s_1^Q, \dots, s_{k^Q}^Q]$, where s_ℓ^Q with $\ell = \{1, \dots, k^Q\}$ is generated from the constellation set $\mathcal{S}^Q$ of the M^Q -ary PAM. p_2^Q bits are used to select another the SAP $\mathcal{I}^Q = \{i_1^Q, \dots, i_{k^Q}^Q\}$ with $\{i_\tau^Q\}_{\tau=1}^{k^Q} \in \{1, 2, \dots, n\}$ for conveying $\mathbf{s}^I$ on the k^Q active subcarriers.

Accordingly, the signal vectors for the in-phase part and the quadrature part can be expressed as

$$\mathbf{x}^I = \frac{1}{\sqrt{I}} [\sqrt{l_1} s_1^I, \sqrt{l_2} s_2^I, 0, \dots, \sqrt{l_{k^I}} s_{k^I}^I, 0, \dots, 0]^T, \quad (52)$$

and

$$\mathbf{x}^Q = \frac{1}{\sqrt{I}} [0, \dots, \sqrt{l_{k^I+1}} s_1^Q, 0, \dots, 0, \sqrt{l_{\alpha}} s_{k^Q}^I]^T, \quad (53)$$

respectively. The transmitted signal vector at the GS is obtained by combining these two vectors, and is given by

$$\mathbf{x} = \mathbf{x}^I + j\mathbf{x}^Q. \quad (54)$$

After propagating over multipath fading channels, the received signal vectors in the frequency domain can be expressed as

$$\mathbf{y}_{u_i}^{t_1} = \mathbf{H}_{s_i} \mathbf{x} + \mathbf{n}_{s_i}^{t_1}, \quad i = 1, \dots, K. \quad (55)$$

Assuming perfect CSI at all receivers, the transmitted signal can be recovered at UAV 1 using the ML detector, and is given by

$$\begin{aligned} & [\hat{\zeta}_{u_1}^{t_1}, (\hat{\mathcal{I}}^I)_{u_1}^{t_1}, (\hat{\mathbf{s}}^I)_{u_1}^{t_1}, (\hat{\mathcal{I}}^Q)_{u_1}^{t_1}, (\hat{\mathbf{s}}^Q)_{u_1}^{t_1}] \\ &= \arg \min_{\zeta, \mathcal{I}^I, \mathbf{s}^I, \mathcal{I}^Q, \mathbf{s}^Q} \|\mathbf{y}_{u_1}^{t_1} - \mathbf{H}_{s_1} \mathbf{x}\|^2, \end{aligned} \quad (56)$$

where $\hat{\zeta}_{u_1}^{t_1}$ is used to recover the information bits of UAV 1, and $\{(\hat{\mathcal{I}}^I)_{u_1}^{t_1}, (\hat{\mathbf{s}}^I)_{u_1}^{t_1}, (\hat{\mathcal{I}}^Q)_{u_1}^{t_1}, (\hat{\mathbf{s}}^Q)_{u_1}^{t_1}\}$ are the estimates of $\{\mathcal{I}^I, \mathbf{s}^I, \mathcal{I}^Q, \mathbf{s}^Q\}$ at UAV 1 in time slot t_1 . The signal detection of UAVs 2 to 5 is performed cluster by cluster in the next phase.

2) *Cooperation Phase*: For $K = 5$, the cooperation phase of MCCU-NOMA-IM requires four time slots, whereas MCCU-NOMA-IM only needs two time slots for the same phase. In time slot t_2 , UAV 1 first reconstructs an IM signal vector $\mathbf{b} = \mathbf{b}^I + j\mathbf{b}^Q$, which is composed of the in-phase part $\mathbf{b}^I$ and the quadrature part $\mathbf{b}^Q$. $\mathbf{b}^I$ is generated by decoding $(\hat{\mathcal{I}}^I)_{u_1}^{t_1}$ and $(\hat{\mathbf{s}}^I)_{u_1}^{t_1}$, resulting in

$$\mathbf{b}^I = [0, \hat{s}_1^I, \dots, 0, \dots, \hat{s}_{k^I}^I]^T. \quad (57)$$

In the same way, $\mathbf{b}^Q$ is generated by decoding $(\hat{\mathcal{I}}^Q)_{u_1}^{t_1}$ and $(\hat{\mathbf{s}}^Q)_{u_1}^{t_1}$ as

$$\mathbf{b}^Q = [\hat{s}_1^Q, 0, \dots, \hat{s}_{k^Q}^Q, 0]^T. \quad (58)$$

Afterward, UAV 1 broadcasts $\mathbf{b}$ to UAV 2 and UAV 3 for cooperation purposes. Meanwhile, UAV 4 and UAV 5 are assumed to be unable to receive the signal from UAV 1 due to severe propagation attenuation caused by the long distance. The received signal vectors at UAV 2 and UAV 3 are given by

$$\mathbf{y}_{u_i}^{t_2} = \mathbf{H}_{u_{i1}} \mathbf{b} + \mathbf{n}_{u_{i1}}^{t_2}, \quad i = 2, 3. \quad (59)$$

After receiving the cooperative vector from UAV 1, the transmitted signals for UAV 2 and UAV 3 can be estimated as

$$\begin{aligned} & [(\hat{\mathcal{I}}^I)_{u_2}^{t_2}, (\hat{\mathbf{s}}^I)_{u_2}^{t_2}] = \arg \min_{\mathcal{I}^I, \mathcal{I}^Q, \mathbf{s}^I, \mathbf{s}^Q} \left\{ \|\mathbf{y}_{u_2}^{t_1} - \mathbf{H}_{s_2} \mathbf{x}\|^2 \right. \\ & \quad \left. + \|\mathbf{y}_{u_2}^{t_2} - \mathbf{H}_{u_{12}} \mathbf{b}\|^2 \right\}, \end{aligned} \quad (60)$$

and

$$\begin{aligned} & [(\hat{\mathcal{I}}^Q)_{u_3}^{t_2}, (\hat{\mathbf{s}}^Q)_{u_3}^{t_2}] = \arg \min_{\mathcal{I}^I, \mathcal{I}^Q, \mathbf{s}^I, \mathbf{s}^Q} \left\{ \|\mathbf{y}_{u_3}^{t_1} - \mathbf{H}_{s_3} \mathbf{x}\|^2 \right. \\ & \quad \left. + \|\mathbf{y}_{u_3}^{t_2} - \mathbf{H}_{u_{13}} \mathbf{b}\|^2 \right\}, \end{aligned} \quad (61)$$

respectively. The information sent to UAV 2 and UAV 3 can be obtained by decoding $(\hat{\mathbf{s}}^I)_{u_2}^{t_2}$ and $(\hat{\mathbf{s}}^Q)_{u_3}^{t_2}$. Subsequently, the estimated SAPs $(\hat{\mathcal{I}}^I)_{u_2}^{t_2}$ and $(\hat{\mathcal{I}}^Q)_{u_3}^{t_2}$ are transmitted to UAV 4 and UAV 5 for further cooperation in the next time slot. Specifically, in time slot t_3 , UAV 2 first generates the SSK vector $\psi = [0, 1, 0, \dots, 1, 0]$ based on $(\hat{\mathcal{I}}^I)_{u_2}^{t_2}$ and transmits it to UAV 4, yielding the received signal as $\mathbf{y}_{u_4}^{t_3} = \mathbf{H}_{u_{24}} \psi + \mathbf{n}_{u_{24}}^{t_3}$. In the meantime, UAV 3 also generates another SSK vector $\chi = [1, 0, \dots, 0, 1]$ based on $(\hat{\mathcal{I}}^Q)_{u_3}^{t_2}$ and transmits it to UAV 5, yielding the received signal as $\mathbf{y}_{u_5}^{t_3} = \mathbf{H}_{u_{35}} \chi + \mathbf{n}_{u_{35}}^{t_3}$.

Finally, the transmitted signals for UAV 4 and UAV 5 can be estimated as

$$(\hat{\mathcal{I}}^I)_{u_4}^{t_3} = \arg \min_{\mathcal{I}^I, \mathcal{I}^Q, \mathbf{s}^I, \mathbf{s}^Q} \left\{ \|\mathbf{y}_{u_4}^{t_1} - \mathbf{H}_{s_4} \mathbf{x}\|^2 + \|\mathbf{y}_{u_4}^{t_3} - \mathbf{H}_{u_{24}} \psi\|^2 \right\}, \quad (62)$$

and

$$(\hat{\mathcal{I}}^Q)_{u_5}^{t_3} = \arg \min_{\mathcal{I}^I, \mathcal{I}^Q, \mathbf{s}^I, \mathbf{s}^Q} \left\{ \|\mathbf{y}_{u_5}^{t_1} - \mathbf{H}_{s_5} \mathbf{x}\|^2 + \|\mathbf{y}_{u_5}^{t_3} - \mathbf{H}_{u_{35}} \chi\|^2 \right\}. \quad (63)$$

The information bits for UAV 4 and UAV 5 can be obtained by decoding $(\hat{\mathcal{I}}^I)_{u_4}^{t_3}$ and $(\hat{\mathcal{I}}^Q)_{u_5}^{t_3}$.

Remark: By comparing the analyses in Section II and Section IV, one can easily observe that for the three-UAV case, MCCU-NOMA-IM requires only two time slots to complete the whole transmission process, whereas MCU-NOMA-IM requires three time slots. Consequently, the end-to-end transmission time for MCCU-NOMA-IM compared with MCU-NOMA-IM is reduced by approximately 33.3%. As the number of UAVs increases, the reduction in end-to-end transmission time becomes more significant. For instance, a 40% reduction in transmission time can be achieved with MCCU-NOMA-IM compared to MCU-NOMA-IM in a network deployment with five UAVs. Based on these results, we can observe that MCCU-NOMA-IM, assisted by a proper grouping strategy, is more suitable for latency-critical applications, especially when the number of UAVs is large.

V. SIMULATION RESULTS

In this section, we present numerical results to evaluate the BER performance of MCU-NOMA-IM and MCCU-NOMA-IM assuming perfect channel estimation, and compare them against benchmarks that do not consider IM and the cooperation phase. For simplicity, PSK is employed as primary modulation scheme. Specifically, we consider the following schemes:

- “MCU-NOMA (n , M PSK)” denotes a multi-UAV cooperative system aided by NOMA (MCU-NOMA) with n subcarriers and employing the M -ary PSK constellation set, which is an application of the cooperative NOMA scheme proposed in [30] for the multi-UAV scenario.
- “MCU-NOMA-IM (n , k , [M_1 PSK, M_2 PSK])” denotes the MCU-NOMA-IM scheme with n subcarriers, k active subcarriers, M_1 -ary PSK constellation set, and distinguishable M_2 -ary PSK constellation set.
- “MCU-NOMA-IM (n , k , I , [M_1 PSK, M_2 PSK])” denotes the MCU-NOMA-IM scheme with n subcarriers, k active subcarriers, total energy coefficient I , M_1 -ary PSK constellation set, and distinguishable M_2 -ary PSK constellation set.
- “MCCU-NOMA-IM (or NC-MU-NOMA-IM) (n , k , [M_1 PSK, M_2 PSK])” denotes the MCCU-NOMA-IM (or non-cooperative multiple UAV-NOMA-IM) scheme with n subcarriers, k active subcarriers, M_1 -ary PSK constellation set, and distinguishable M_2 -ary PSK constellation set, where NC-MU-NOMA-IM is a generalization of the NOMA-IM scheme proposed in [34] for the multi-clustered-UAV scenario.

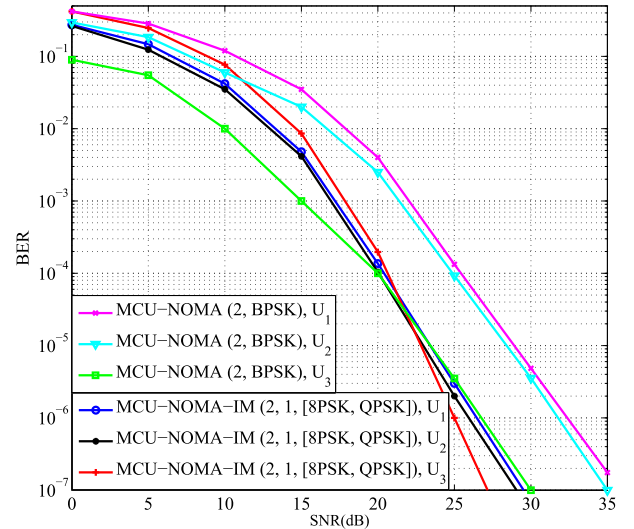


Fig. 3. BER comparison between MCU-NOMA-IM and MCU-NOMA with $n = 2$, $k = 1$, and three UAVs by setting the SE to 3 bpcu, where the power coefficients of the MCU-NOMA scheme are 0.05, 0.15, and 0.8, respectively, corresponding to UAVs 1 ~ 3, and m_Δ is set to 3 for $\Delta \in \{s_i, u_{ij}\}$ with $i, j \in \{1, 2, 3\}$.

- “MCCU-NOMA-IM (or NC-MU-NOMA-IM) (n , k_1 , k_2 , I , [M_1 PAM, M_2 PAM])” denotes the MCCU-NOMA-IM (or NC-MU-NOMA-IM) scheme with n subcarriers, k_1 (k_2) active subcarriers for the in-phase (quadrature) part, total energy coefficient I , and M_1 (M_2)-ary PAM constellation set for the in-phase (quadrature) part.

In Fig. 3, we compare the BER of MCU-NOMA-IM and MCU-NOMA when $n = 2$ and $k = 1$ for three UAVs, and the SE is 3 bit per channel use (bpcu). It can be seen from Fig. 3 that the UAVs using MCU-NOMA-IM achieve better BER performance than those using MCU-NOMA. Specifically, compared to “MCU-NOMA (2, BPSK)”, UAV 1 and UAV 2 using “MCU-NOMA-IM (2, 1, [8PSK, QPSK])” have better BER performance over the considered SNR range, achieving about 7 dB SNR gains at $\text{BER} = 10^{-6}$. The significant SNR gains for UAV 1 and UAV 2 are harvested thanks to the use of interference-free and orthogonal signal dimensions, without requiring the SIC-based detection among all the UAVs. More interestingly, UAV 3 using “MCU-NOMA-IM (2, 1, [8PSK, QPSK])” has worse performance than UAV 3 using “MCU-NOMA (2, BPSK)” in the low SNR region ($\text{SNR} < 22$ dB), while outperforming it as the SNR increases ($\text{SNR} > 22$ dB). Except for the non-interference transmission, the performance improvement of UAV 3 using “MCU-NOMA-IM (2, 1, [8PSK, QPSK])” can be mainly attributed to the increased diversity order introduced by IM. In addition, we observe that UAV 1 and UAV 2 using “MCU-NOMA-IM (2, 1, [8PSK, QPSK])” as well as all the UAVs using “MCU-NOMA (2, BPSK)” achieve the same diversity order through the transmission of modulation bits, while only UAV 3 using “MCU-NOMA-IM (2, 1, [8PSK, QPSK])” achieves a higher diversity order because of the transmission of additional index bits.

To further examine the advantages of the proposed scheme, in Fig. 4, we compare the BER of MCU-NOMA-IM and MCU-NOMA when $n = 4$, $k = 2$, and $I = 5$ for four UAVs by assuming that the SE is 4 bpcu. One can observe

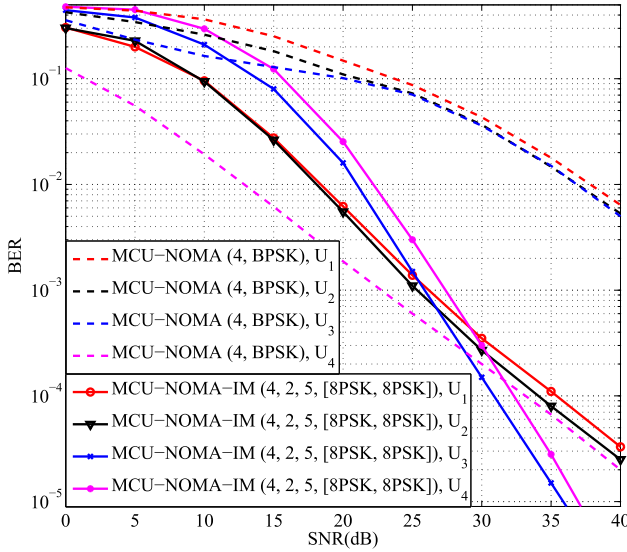


Fig. 4. BER comparison between MCU-NOMA-IM and MCU-NOMA with $n = 4$, $k = 2$, $I = 5$, and four UAVs by setting the SE to 4 bpcu, where the power coefficients of the MCU-NOMA scheme are 0.01, 0.04, 0.1, and 0.85, respectively, corresponding to UAVs 1 $\sim$ 4, and m_Δ is set to 1 for $\Delta \in \{s_i, u_{ij}\}$ with $i, j \in \{1, 2, 3, 4\}$.

that, although the BER of distant UAVs using MCU-NOMA is excellent, the BER of near UAVs is degraded due to the intricate power allocation design, especially when the number of UAVs is large. In this case, improving the BER of near UAVs by MCU-NOMA is crucial. From Fig. 4, it becomes apparent that the BER of UAVs 1, 2, and 3 (all near UAVs) using “MCU-NOMA (4, BPSK)” is considerably worse, while UAVs 1, 2, and 3 utilizing “MCU-NOMA-IM (4, 2, 5, [8PSK, 8PSK])” have a lower BER by using IM. Moreover, comparing the BER of UAV 4 using MCU-NOMA and MCU-NOMA-IM, we see an intersection of the corresponding curves. Specifically, when $\text{SNR} > 30$ dB, UAV 4 using the proposed scheme can achieve better performance due to the higher diversity resulting from IM, but is also experiences a performance loss when $\text{SNR} < 30$ dB. Additionally, we can observe that UAV 3 and UAV 4 using “MCU-NOMA-IM (4, 2, 5, [8PSK, 8PSK])” present the same diversity order, since they use the EAP and SAP for data modulation, respectively.

To verify the theoretical analysis, we compare the BER of MCU-NOMA-IM with three and four UAVs in Fig. 5. In Fig. 5(a), we analyze the BER of “MCU-NOMA-IM (4, 2, [BPSK, BPSK])” and see that the curves obtained by using the analytical framework exhibit a slight mismatch with the numerical simulations in the low SNR region, but they are sufficiently accurate when $\text{SNR} \geq 30$ dB. Moreover, we note that the theoretical analysis of UAV 1 and UAV 2 using MCU-NOMA-IM with three and four UAVs is the same. Therefore, we only need to consider the performance of UAV 3 and UAV 4 for “MCU-NOMA-IM (4, 2, 5, [BPSK, BPSK])” in Fig. 5(b) when the number of UAVs is four. From Fig. 5(b), we can find that the theoretical curves of UAV 3 and UAV 4 tightly match the simulation results when $\text{SNR} \geq 30$ dB. Accordingly, the obtained numerical results validate the accuracy of the performance analysis given in Section III.

To evaluate the BER of MCCU-NOMA-IM, we illustrate numerical results when $n = 4$ and $k = 2$ in Fig. 6.

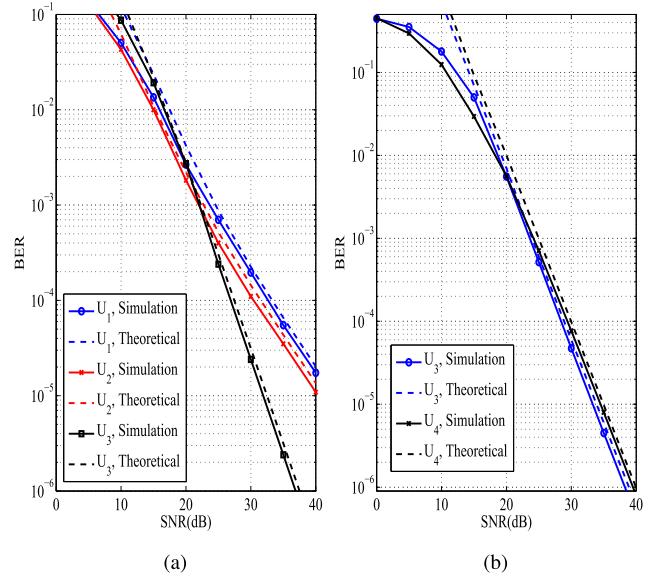


Fig. 5. Performance comparison between simulation and theoretical results for the MCU-NOMA-IM scheme, where m_Δ is set to 1 for $\Delta \in \{s_i, u_{ij}\}$: (a) $n = 4$, $k = 2$, $I = 5$, BPSK, and three UAVs; (b) $n = 4$, $k = 2$, $I = 5$, BPSK, and four UAVs.

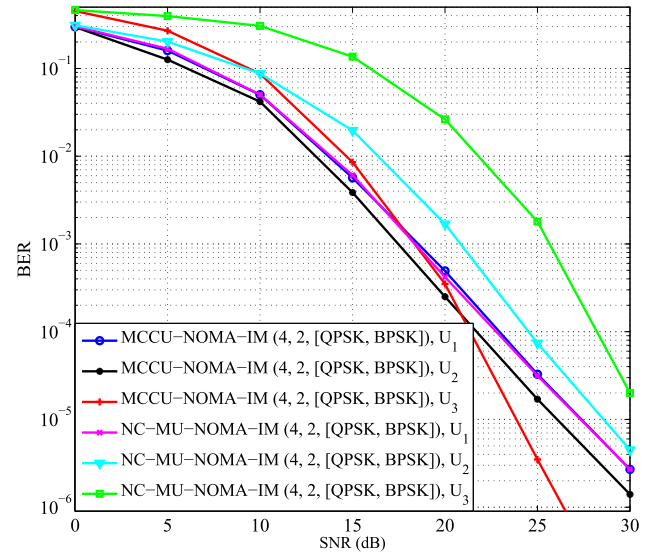


Fig. 6. BER comparison between MCCU-NOMA-IM and NC-MU-NOMA-IM with $n = 4$, $k = 2$, and three UAVs, where m_Δ is equal to 2 for $\Delta \in \{s_i, u_{ij}\}$ with $i, j \in \{1, 2, 3\}$.

Specifically, it can be seen that UAV 1 achieves the same BER using “MCCU-NOMA-IM (4, 2, [QPSK, BPSK])” and “NC-MU-NOMA-IM (4, 2, [QPSK, BPSK])”, since UAV 1 does not get any support from other UAVs in both schemes. In contrast, UAV 2 and UAV 3 using “MCCU-NOMA-IM (4, 2, [QPSK, BPSK])” achieve about 3 dB and 8 dB SNR gains at $\text{BER} = 10^{-5}$ compared to their counterparts using “NC-MU-NOMA-IM (4, 2, [QPSK, BPSK])”. These results unveil that the cooperation is more efficient for distant UAVs, especially for UAV 3 that utilizes index bits. This is because the channel condition becomes statistically worse as the distance increases, which degrades the BER, and the cooperation from near UAVs can partially compensate for the performance degradation. It is worth noting, however, that the substantial

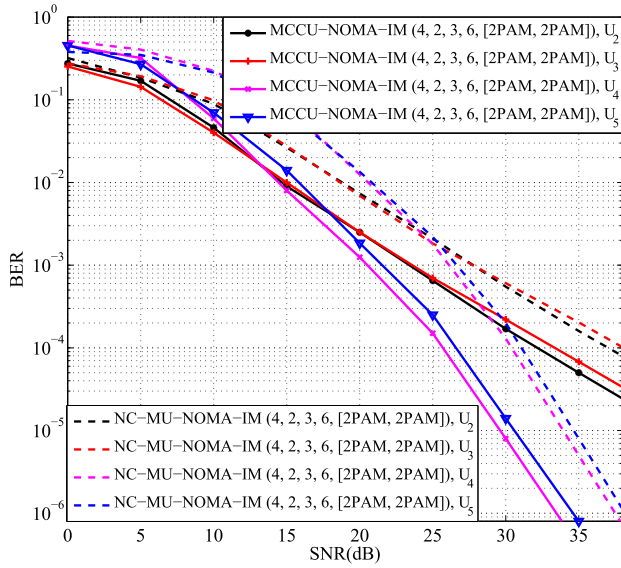


Fig. 7. BER comparison between MCCU-NOMA-IM and NC-MU-NOMA-IM with $n = 4$, $k = 2$, $I = 6$, and five UAVs, where m_{Δ} is equal to 1 for $\Delta \in \{s_i, u_{ij}\}$ with $i, j \in \{1, 2, 3, 4, 5\}$.

performance gain brought by the proposed MCCU-NOMA-IM scheme is obtained at the cost of a longer transmission latency.

To further examine the efficacy of MCCU-NOMA-IM, we compare the BER of MCCU-NOMA-IM and NC-MU-NOMA-IM when $n = 4$, $k_1 = 2$, $k_2 = 3$, $I = 6$, and 2PAM for five UAVs in Fig. 7. Since the BER of UAV 1 using the MCCU-NOMA-IM and NC-MU-NOMA-IM schemes are the same, we do not illustrate the curves of UAV 1 for simplicity. From Fig. 7, we can see that UAVs 2, 3, 4, and 5 using “MCCU-NOMA-IM (4, 2, 3, 6, [2PAM, 2PAM])” can still achieve a considerable performance improvement compared to the “NC-MU-NOMA-IM (4, 2, 3, 6, [2PAM, 2PAM])” scheme. Interestingly, we find that UAV 2 and UAV 3 grouped in one cluster achieve similar BER in terms of modulation bits. A similar phenomenon can also be observed for UAV 4 and UAV 5 grouped in another cluster, ensuring BER fairness among UAVs within a cluster. This is because all UAVs grouped in the same cluster are located at a similar distance from the GS, and the corresponding signal is conveyed via independent in-phase and quadrature components, which validates the efficacy of the proposed clustering design for MCCU-NOMA-IM. In summary, it is worthwhile trading additional time slots for harvesting significant performance gains by MCCU-NOMA-IM.

VI. CONCLUSION

In this paper, we proposed the MCU-NOMA-IM scheme to explore the potential of IM to air-ground networks consisting of multiple UAVs. By virtue of multiple independent transmission dimensions, the UAVs are able to receive interference-free signals in the downlink, which greatly improves the overall system performance. Moreover, upper bounds for the BER of MCU-NOMA-IM were derived in closed-form. In order to reduce the end-to-end transmission latency when applying

MCU-NOMA-IM, we proposed an enhanced scheme, named MCCU-NOMA-IM, by grouping nearby UAVs into clusters. By doing so, compared to NC-MU-NOMA-IM, MCCU-NOMA-IM achieves lower error performance, especially when the number of UAVs is large. Finally, extensive analytical and simulation results showed that MCU-NOMA-IM and MCCU-NOMA-IM are superior to MCU-NOMA and NC-MU-NOMA-IM, and validated the accuracy of the theoretical analysis of the proposed schemes.

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